

B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level							
Sem:		II					
Paper Category:		Practical (Major) (Group-II)					
Paper Name:		Practical based on DSC4-2			Paper Code : G06-0204-P		
Credit:		02		Practical:		4 Hrs./Week	
Marks:	UA:	30	CA:	20	Total:	50	

Choose one of the following languages to solve the assignment: Python, R, or MATLAB.

1. Implement functions to check if one set is a subset of another.
2. Write functions to generate the power set of a set using iterative or recursive methods.
3. Create functions to perform union, intersection, complement, set difference and symmetric difference between of two sets.
4. Implement functions to verify Commutative, Distributive, and Associative Laws properties for set operations.
5. Write functions to validate DeMorgan's laws for sets.
6. Develop algorithms to determine if two sets are disjoint.
7. Implement a function to calculate the cardinality (number of elements) of a set.
8. Write functions to compute the Cartesian product of two sets.
9. Implement functions to classify relations is Void, universal, identity, reflexive, symmetric, transitive, Equivalence, antisymmetric, partial ordering, and asymmetric.
10. Implement functions for matrix operations like addition and multiplication specific to relation matrices.
11. Use Matplotlib or similar libraries to visualize relations as directed graphs (digraphs).
12. Write functions to Calculate indegree and outdegree for vertices in a digraph.
13. Implement Warshall's algorithm to find the transitive closure of a relation.
14. Implement examples to illustrate the pigeonhole principle in counting.
15. Write functions to apply the addition and multiplication principle for counting sets.
16. Develop functions to solve problems using inclusive-exclusive principles for two sets and three sets.
17. Solve advanced problems involving combinations, permutations, and constraints using counting principles.